

Eric Nieto Gonzalez

LinkedIn: www.linkedin.com/in/ericnietogonzalez || **Phone Number:** 312-978-1993
Personal Website: <https://rickstarr1.github.io/Personal-Website/> || **Email:** nietoeric1@gmail.com

Education:

University of Illinois at Urbana-Champaign

- M.S. in Electrical Engineering
 - GPA: 4.0 / 4.0
- B.S. in Electrical Engineering
 - GPA: 3.55 / 4.0

Grainger College of Engineering
Expected Graduation: May 2027

August 2020 → May 2025

Work Experience:

Research Intern at Sandia National Laboratories

- Used Nanoscribe Quantum X 2PP to fabricate 3D nanostructures with <200 nm resolution
- Operated E-beam evaporator (Temescal FCS2000) to deposit metals with <5 nm thickness
- Utilized SEM for nanostructure characterization and process validation
- Trained in cleanroom wet bench protocols (acid, solvent, base)

May 2025 → August 2025

Soft Robotics Research Assistant at the University of Illinois Urbana-Champaign

- Created a microgripper with <10 μm diameter for potential human vein procedures and studies
- Applied 2.5D Topology optimization to create an efficient gripper with minimal material
- Developed 3 alternate innovative medical microtools to enhance versatility in applications

January 2024 → May 2025

Manufacturing Engineering Intern at G&W Electric Co.

- Designed custom tools that improved manufacturing efficiency by 40%
- Proficient in industry engineering documentation, including product and electrical schematics
- Coordinated with 12 employees across all specialties to collect input for project development

May 2024 → August 2024

Leadership Roles:

Teaching Assistant for ECE 340: Semiconductor Devices

- Created 1 original problem per assignment to reinforcing semiconductor concepts
- Facilitated weekly office hours, providing individual support to 10 students per session
- Evaluated student performance by grading homework and exams through constructive feedback

August 2025 → Present

Society of Hispanic Professional Engineers (SHPE)

- Planned and hosted social events for 50 undergraduates within engineering to create connections
- Communicated details of all social events thoroughly to all target groups in an effective manner

January 2022 → Present

Projects:

Nanoscale Semiconductor Devices and Simulation

- Studied nanoscale semiconductor properties and carbon nanomaterials
- Gained hands-on experience with nanofabrication techniques within a cleanroom

January 2025 → May 2025

Advanced Optoelectronic Device Characterization

- Explored energy conversion devices for LEDs and solar cells
- Analyzed heterostructures and low-dimensional quantum structures for modern devices
- Characterized LEDs and solar cells using TCAD simulations for electrical and optical modeling

January 2025 → May 2025

Integrated Circuit Wafer Fabrication and Testing

- Fabricated semiconductor devices using photolithography, diffusion, and etching techniques
- Developed cleanroom skills to produce functioning MOS transistors and diodes
- Conducted device testing to ensure functionality, identify defects, and validate the design.
- Analyzed process parameters to refine microfabrication techniques and improve device yield.

August 2024 → December 2024

Developed SNES Street Fighter 2 using System Verilog

- Coded modules in System Verilog which handle multiple sections of the game simultaneously
- Numerous modules created to make it more efficient to debug, understand, and improve
- Drivers integrated to receive keyboard signals and send out VGA signals within the board
- Wrote a thorough, concise, and formal lab report relating to the specific details of the game

April 2023 → May 2023

Skills:

- PCB Design, KiCad
- System Verilog, HDL
- C Language
- MATLAB
- PTC Creo / SolidWorks
- Seal of Bilingualism in Spanish & English

Awards:

- UIUC Knight of St. Patrick's Awardee
- Intel Scholar
- ILLCF Scholarship Recipient
- Hispanic Scholarship Fund Recipient
- HRP Scholarship Recipient
- Wentcher Scholarship Recipient